

## BASIC NETWORKING COMMANDS

Aim:-

Study of various Network commands used in Linux and Windows.

- 1) `arp -a` : It shows the computer's IP address and the IP and MAC addresses of devices on your local network.

O/p:-

Interface : 192.168.56.1 --- 0x7

| Internet address | Physical address  | Type   |
|------------------|-------------------|--------|
| 192.168.56.255   | ff-ff-ff-ff-ff-ff | Static |
| 224.0.0.2        | 01-00-5e-00-00-02 | Static |

- 2) `hostname` - It is the simplest of all TCP/IP commands. It simply displays the name of your computer.

O/p:- hp-Kaviya-E

- 3) `ipconfig/all` - It displays detailed configuration information about your TCP/IP connection including Router, Gateway, DNS, DHCP, and type of Ethernet adapter in your system.

O/p:-

Windows IP Configuration

Host name . . . . . : hp-Kaviya-E

Primary DNS suffix :

Nbde Type . . . . . : Mixed

IP Routing Enabled : . . . . : No

WINS Proxy Enabled : No.



4) `nbtstat -a`: This command helps solve problems with NetBIOS name resolution.

O/p:- Displays Protocol Statistics and current TCP/IP connections using NBT.

-a (adapter status) Lists the remote machine's name table given its name.

-A (Adapter status) Lists the remote machine's name table given its IP address.

-c (cache) Lists NBT's cache of remote names and their IP address.

-n (names) List's local NetBIOS names.

5) `netstat` - Displays active network connections, routing tables, and network statistics.

| Proto | Local Address   | Foreign Address   | State       |
|-------|-----------------|-------------------|-------------|
| TCP   | 127.0.0.1:49678 | hp-Kaviya-E:49679 | Established |
| TCP   | 127.0.0.1:49679 | hp-Kaviya-E:49678 | Established |
| TCP   | 127.0.0.1:46680 | hp-Kaviya-E:49681 | Established |
| TCP   | 127.0.0.1:46681 | hp-Kaviya-E:49680 | Established |
| TCP   | 127.0.0.1:46705 | hp-Kaviya-E:49686 | Established |

6) `nslookup` - Used to find DNS information, such as the IP address of a domain and DNS records.

O/p: Default server: Unknown.

Address: 172.16.8.1

7) `pathping`: Is unique to Windows and combines the ping and Tracert commands. It traces the route to the destination and gathers statistics on data loss at each hop.



O/p:-

Usage: pathping [-g host\_list] [-h maximum\_hops]  
[-i address] [-n] [-p period] [-q num-  
queries] [-w timeout] [-4] [-6]  
target\_name.

-g host\_list Loosely source route along host\_list  
-h maximum\_hops Max no. of hops to search for  
target.

8) ping - Is best way to test connectivity between two  
nodes. Ping use ICMP to communicate to other  
devices.

O/p:- ping [-t] [-a] [-n count] [-I size] [-f] [-i TTL]  
[-v TOS] [-r count] [-s count] [-j host\_list] |  
[-k host\_list] [-w timeout] [-R] [-S srcaddr]  
[-c Compartment] [-p] [-4] [-6] target\_name.

9) Route - Used to manipulate the IP routing table.  
It is primarily used to setup static routes to specific  
host or networks via an interface.

O/p:- ROUTE [-f] [-p] [-4] [-6] Command [destination]  
[MASK netmask] [gateway] [METRIC metric]  
[IF interface].



# LINUX COMMANDS:-

1) **ip** - The **ip** command is used to manage IP addresses, network interfaces, and routes. It also helps troubleshoot network connectivity.

O/e:- **ip [OPTIONS] OBJECT {COMMAND} help**  
**ip [-force] -batch filename**

② to show ip address assigned to an interface on your server.

**ip address show.**

1:10::< LOOP BACK UP, LOWER UP > m 65536

⑥ **ip address add 192.168.1.254/24 dev enp03**

⑦ **ip address del 192.168.1.254/24 dev enp03.**

2) **ipconfig** : It was a staple in many sysadmin tool ~~set~~ for configuring and troubleshooting networks. It has since been replaced by the **ip** command discussed above.

O/p:- enp03 flags 0183<OP, BROADCAST RUNNING  
MULTICAST 1500 init 10.0.2.15 netmask  
255.55 255.2 255.0

3) **mttr** : MTR is a program with a command line interface that serves as a network diagnostic and troubleshooting tool. This command combines the functionality of the **ping** and **traceroute** commands.

O/p:- my traceroute [w. 94] I

| host      | Packets              | pings              |
|-----------|----------------------|--------------------|
| localhost | Loss %: 0.0% 122 0.1 | Avg Best worst Dev |
|           |                      | 0.2 0.1 1.9 0.2    |



4) tcpdump:- The tcpdump command is designed for capturing and displaying packets.

0/p'r

tcpdump: you don't have permission to capture on that device.

5) Ping:- ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending Internet Control Message.

• Ping google.com

150

Pinça google com (142.200.182.78) 56 (84)

bytes of data 64 bytes from lemma - an-  
~~in-  
 t~~

1e100.net (142.250.182.78): Emp-Seg

$= 1 \text{ } t + t_i = 112 \text{ time} = 988 \text{ ms.}$

8/11

Result:-

Thus the basic networking and Linux commands are studied and learned, the output got verified.